



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2025

BIOLOGY (N530)

TIME: 45 MINUTES

Candidates answer on the Question Paper.

Additional Materials: Ruler, Calculator

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Diagrams are not drawn to scale unless otherwise specified.
5. Any rough working should be done in this booklet.
6. Answer **ALL** questions.
7. This document consists of **6** questions printed on **12** pages, numbered **2** to **13**.
8. The number of marks is given at the end of each question or part question.
9. The total number of marks for this paper is **50**.

For Examiners' use								
Question No.	1	2	3	4	5	6	Total	Signature
Examiner								
Team Leader								
CE/ACE								

Question 1 (10 marks)

Circle the correct answer. Each item carries one mark.

(a) Which of the following is a function of plasma in blood?

- A** It produces oxygen.
- B** It produces antibodies.
- C** It acts as a barrier to germs.
- D** It transports carbon dioxide.

(b) **Fig. 1.1** shows an artery.

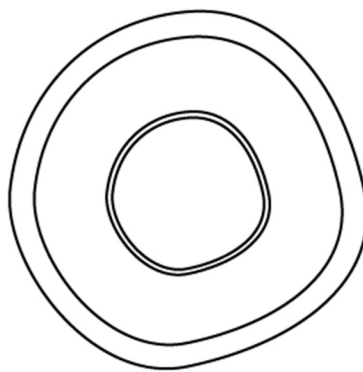


Fig. 1.1: An artery

Which of the following statements about an artery is true?

- A** It carries blood rich in carbon dioxide.
- B** It carries blood under low pressure.
- C** It carries blood under high pressure.
- D** It carries blood towards the heart.

(c) **Fig. 1.2** shows a sample of human blood.

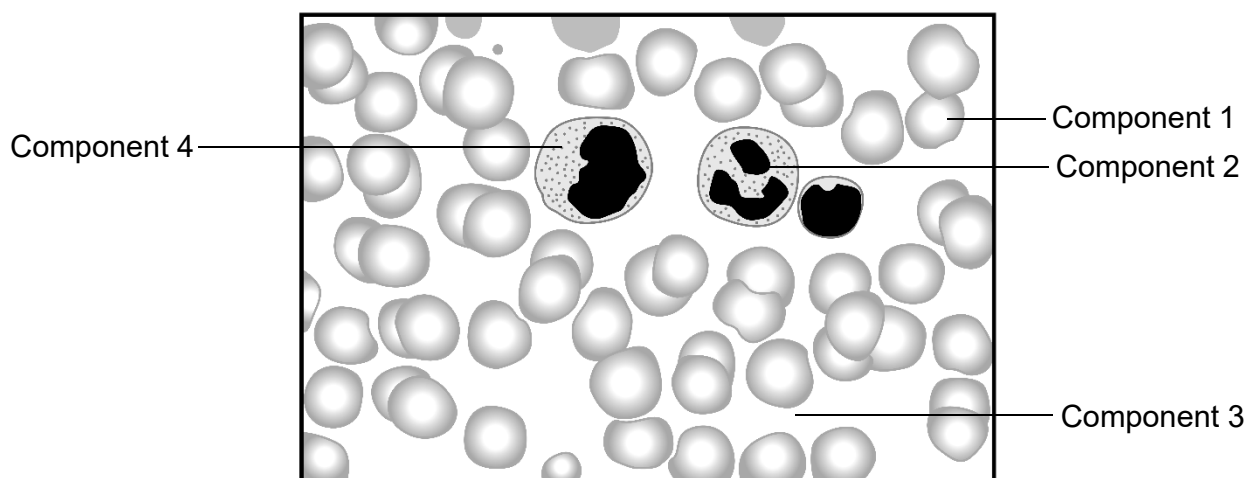


Fig. 1.2: Sample of human blood

In which component is haemoglobin found?

- A** Component 1
- B** Component 2
- C** Component 3
- D** Component 4

(d) Which of the following statements about sexual reproduction is true?

- A** It involves the formation of identical offspring.
- B** It involves a single parent.
- C** It involves the formation of gametes.
- D** It involves the division of a cell into two.

- (e) **Fig. 1.3** shows a side view of the female reproductive system.

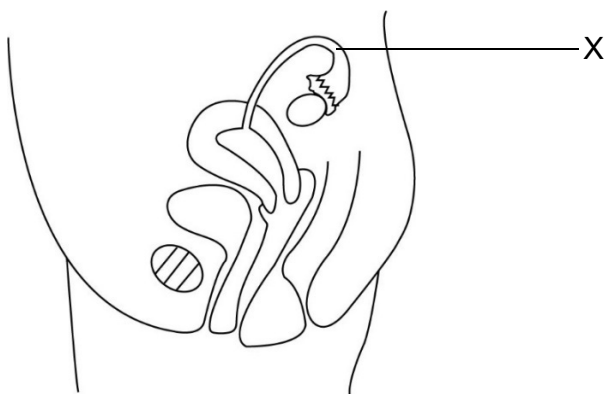


Fig. 1.3: Side view of a female reproductive system

What is part X?

- A** Ovary
 - B** Oviduct
 - C** Uterus
 - D** Vagina
- (f) Which organ of the male reproductive system produces sperms?
- A** Penis
 - B** Scrotum
 - C** Testis
 - D** Urethra
- (g) Which of the following causes 'AIDS'?
- A** A virus
 - B** A fungus
 - C** A bacterium
 - D** A mosquito

(h) Which of the following is produced in plants during photosynthesis?

- A** Chlorophyll
- B** Glucose
- C** Minerals
- D** Water

(i) Which of the following can be counted using a quadrat?

- A** Snails
- B** Butterflies
- C** Frogs
- D** Trees

(j) **Fig. 1.4** shows the structure of a leaf.

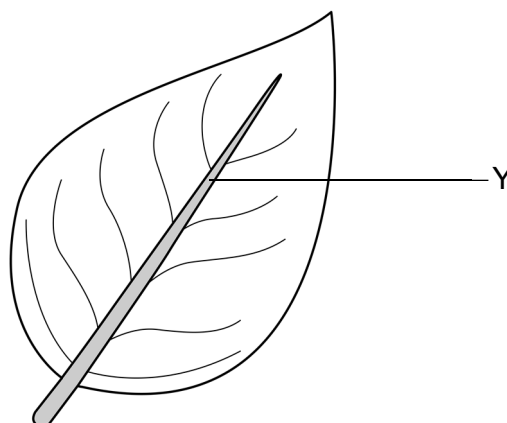


Fig. 1.4: Structure of a leaf

What does Y represent?

- A** Leaf apex
- B** Leaf margin
- C** Midrib
- D** Network of veins

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Question 2 (7 marks)

- (a) Match each part of a leaf in **Column A** with its function in **Column B**.

Column A		Column B
Phloem	•	• To trap light energy from the sun
Xylem	•	• To allow gas exchange
Stomata	•	• To transport water and minerals
Chlorophyll	•	• To maximise absorption of light
		• To transport food

[4]

- (b) **Fig. 2.1** illustrates the process of photosynthesis.

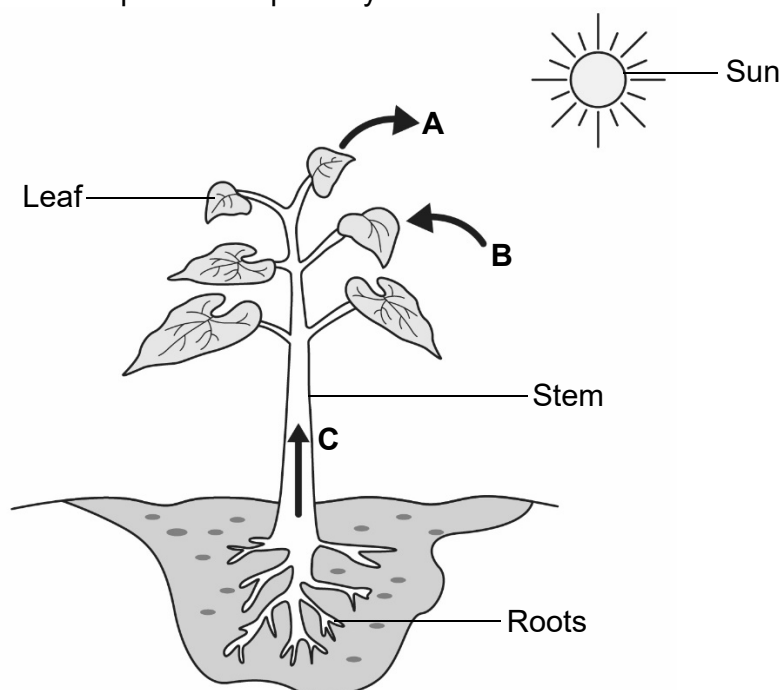


Fig. 2.1: Photosynthesis

In the process of photosynthesis, what do arrows **A**, **B** and **C** represent?

A:

B:

C:

[3]

Question 3 (7 marks)

(a) Give one socio-economic and one ecological benefit of biodiversity.

(i) Socio-economic benefit:

.....

(ii) Ecological benefit:

.....

[2]

(b) (i) Give one example of a **natural calamity** that affects biodiversity.

.....

[1]

(ii) Give one way in which natural calamities affect biodiversity.

..... [1]

(c) (i) State one possible **human activity** that leads to the decline of forest areas.

.....

[1]

(ii) Give two other examples of human activities that affect biodiversity.

1.

2.

[2]

Question 4 (8 marks)

- (a) **Fig. 4.1** shows an organ.
Fig. 4.2 shows part of the human body.

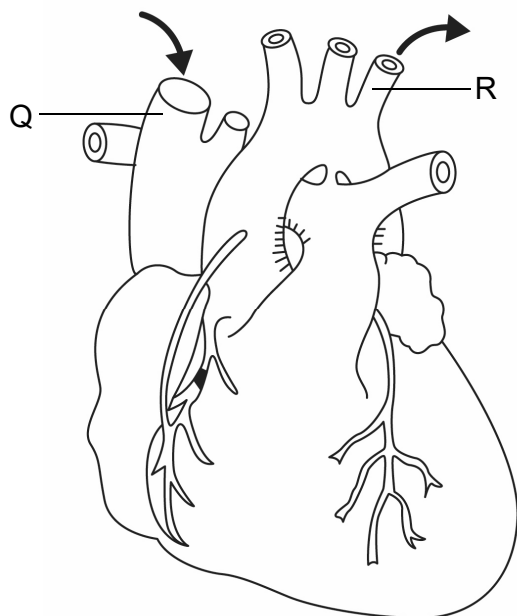


Fig. 4.1

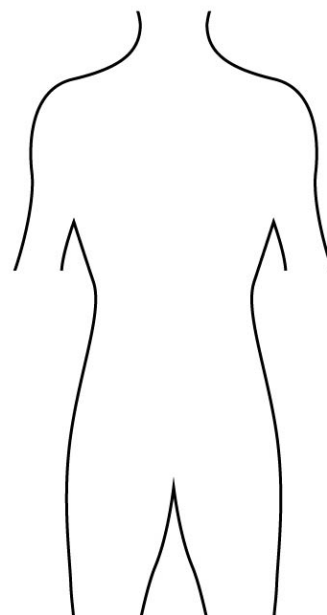


Fig. 4.2

- (i) Name the organ shown in **Fig. 4.1**.

.....

[1]

- (ii) What is the function of this organ in the body?

.....

..... [1]

- (iii) On **Fig. 4.2**, indicate with an 'X' the location of the organ shown in **Fig. 4.1**.

[1]

- (iv) The arrows shown on **Fig. 4.1** indicate the direction of blood flow.

What **type** of blood vessels are Q and R? Choose from the list of words given.

artery

capillary

vein

Q:

[1]

R:

[1]

(b) **Fig. 4.3** shows a section through a blocked coronary artery.

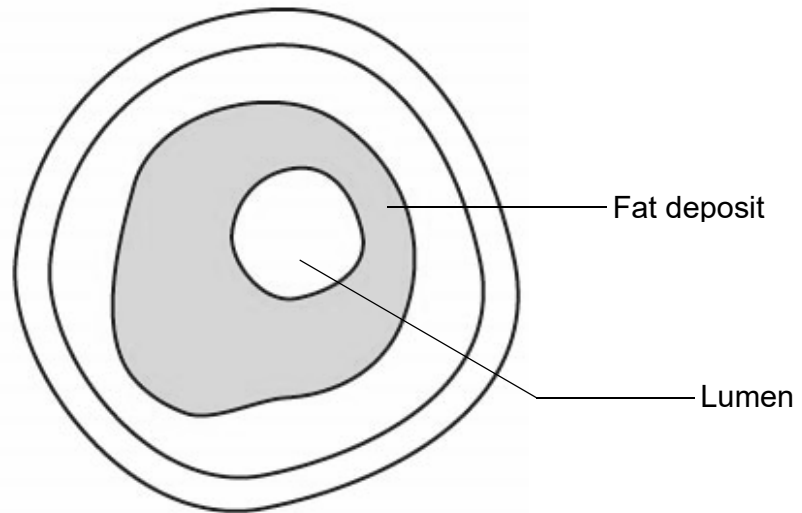


Fig. 4.3: A blocked coronary artery

(i) Name one cardiovascular disease caused by blocked coronary arteries.

.....

[1]

(ii) Give **two** other possible causes of cardiovascular diseases.

1.

2.

[2]

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Question 5 (7 marks)

A student carries an experiment on photosynthesis as follows:

1. He starts the experiment by leaving a plant in darkness for 48 hours.
2. After 48 hours, he covers part of a green leaf of the plant with a black paper strip.
3. He leaves the plant in sunlight, as shown in **Fig. 5.1**.
4. After a few hours in sunlight, he removes the partially covered leaf from the plant and tests it for the presence of starch.

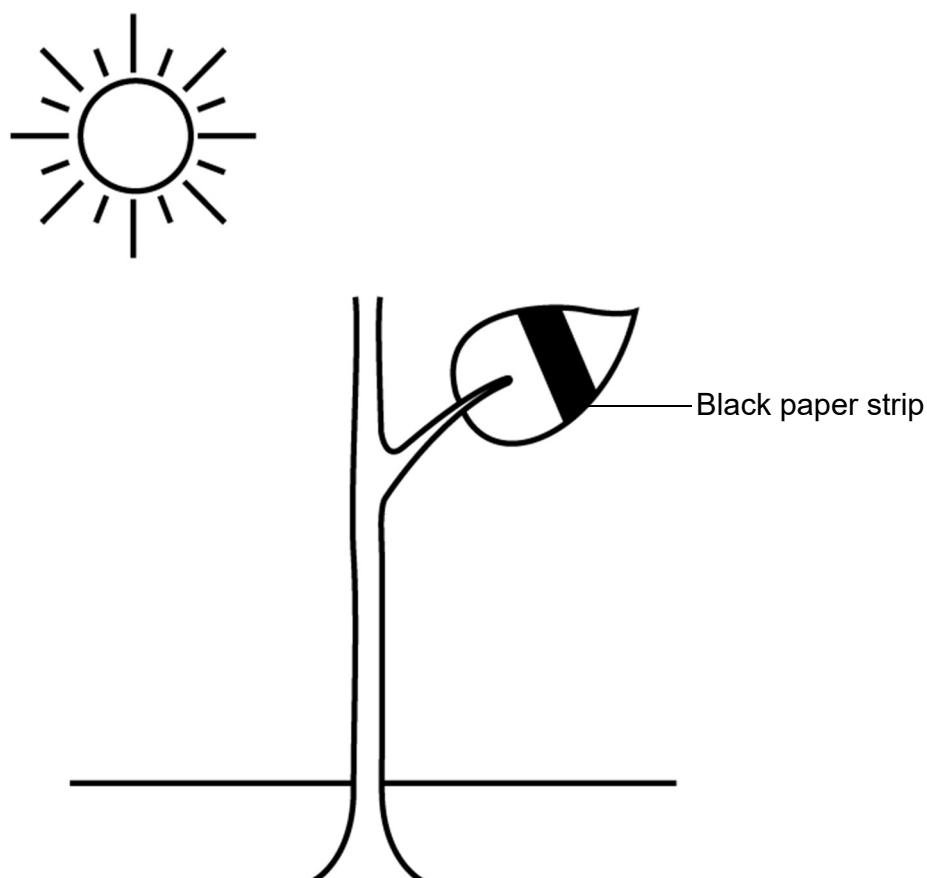


Fig. 5.1: Green leaf partially covered with black paper strip

- (a) Suggest why the student starts the experiment by leaving the plant in darkness for 48 hours.

..... [1]

- (b) Before testing the leaf for starch, the student removes the black paper strip from the leaf.

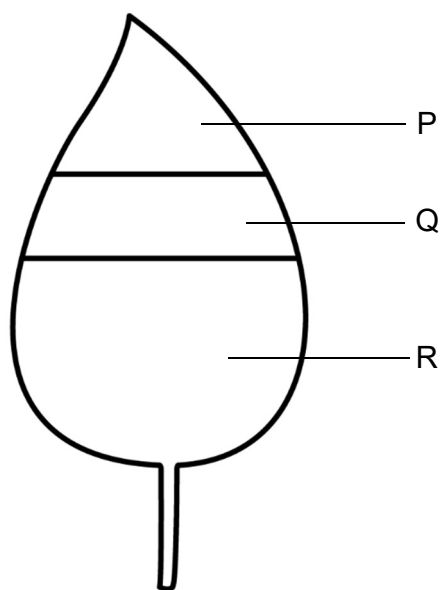


Fig. 5.2

When tested for starch, which **two** labelled parts of the leaf in **Fig. 5.2** will show a blue-black colour?

..... and [2]

- (c) List, in the correct order, the steps to test a leaf for starch.
The first one has been done for you.

1. Put the leaf in boiling water.
2.
3.
4. [3]

- (d) What conclusion can be drawn from this experiment?

..... [1]

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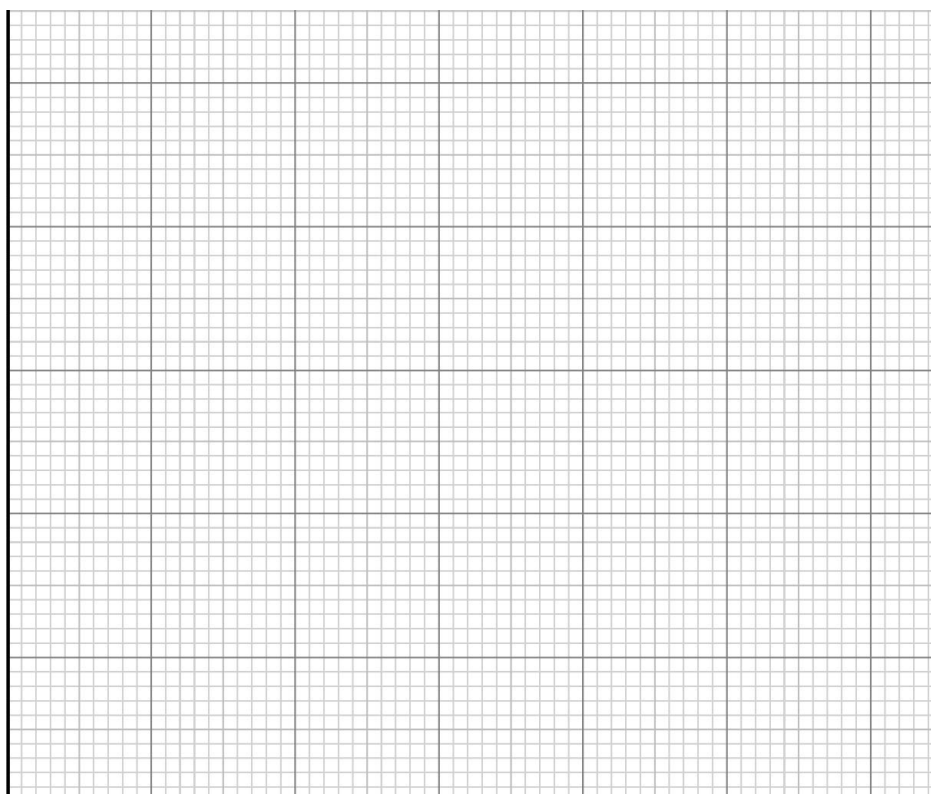
Question 6 (11 marks)

The total number of new cases of HIV/AIDS reported in Mauritius from 2017 to 2022 is given in **Table 6.1**.

Table 6.1

Year	Total number of new cases of HIV/AIDS
2017	380
2018	410
2019	390
2020	340
2021	350
2022	420

- (a) Using the data in **Table 6.1**, plot a graph on the grid provided below. Draw a line to connect each plotted point.



[4]

- (b) Describe the trend of new HIV/AIDS cases reported in Mauritius from 2017 to 2022.

.....

 [2]

- (c) Calculate the average number of new HIV/AIDS cases reported from 2017 to 2022. Show your working.

Average number of new HIV/AIDS cases reported = [2]

- (d) Suggest a preventive measure that may help to reduce the number of new HIV/AIDS cases.

.....
 [1]

- (e) Name two other sexually transmitted diseases.

1.
 2.

[2]

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